

Notes: Fuster, Plosser, Schnabl, and Vickery (RFS, 2019)

The Role of Technology in Mortgage Lending

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1 Big picture

- **Question.** Does financial technology improve mortgage lending, or is FinTech mortgage growth mainly driven by marketing, regulatory arbitrage, or selection?
- **Definition of FinTech lender.** The paper classifies a mortgage originator as FinTech if the application process can be completed entirely online, supported by centralized operations rather than local branches and brokers.
- **Empirical setting.** The paper studies the U.S. mortgage market from 2010 to 2016, using loan-level HMDA application/origination data, Ginnie Mae FHA mortgage performance data, and CRISM mortgage servicing/credit bureau data.
- **Market-share fact.** FinTech lenders grew from about 2% of U.S. mortgage originations in 2010 to about 8% in 2016, with especially strong growth in refinancing and FHA segments.
- **Main processing-time result.** FinTech lenders process mortgage applications faster. The estimated advantage is about 7-10 days for purchase loans and about 10-15 days for refinance loans, or roughly 20% of average processing time.
- **Screening/default result.** Faster processing does not appear to come from laxer screening. FHA loans originated by FinTech lenders have lower, not higher, subsequent delinquency/default rates in the main specifications.
- **Scalability result.** FinTech lenders are less sensitive to aggregate mortgage demand surges. When application volume rises, non-FinTech processing times increase more than FinTech processing times.
- **Denial-rate result.** FinTech lenders do not respond to application-volume shocks by rationing credit more aggressively; their denial probabilities fall differentially when volume rises.
- **Refinancing result.** Areas with higher FinTech mortgage market share refinance more, especially when refinancing is economically sensible according to the Agarwal-Driscoll-Laibson rule.
- **Access-to-finance result.** The authors do not find strong evidence that FinTech lenders primarily expand credit to underserved borrowers or borrowers with low traditional access to finance.
- **Economic takeaway.** Technology mainly reduces transaction and processing frictions in mortgage lending. The benefits are faster origination, more elastic supply, and improved refinancing efficiency rather than broad expansion of credit access.

2 Data and measurement

2.1 FinTech lender classification

- The authors classify lenders by manually examining whether the mortgage application can be completed fully online.
- The classification is lender-level and is applied to the HMDA lender identifiers.
- FinTech lenders in the sample are mostly nonbank mortgage originators that securitize loans and do not rely on branch networks or deposit funding.
- The largest FinTech lenders in 2016 include Quicken Loans, LoanDepot, and Guaranteed Rate.

Interpretation: The classification focuses on the business-process technology used to originate the mortgage, not simply whether the lender has a website. The paper is therefore about online origination technology and centralized processing.

2.2 HMDA mortgage application and origination data

- The Home Mortgage Disclosure Act data cover mortgage applications and originations by lender, borrower, geography, loan purpose, and loan characteristics.
- HMDA includes application outcomes: originated, approved but not accepted, denied, withdrawn, and closed for incompleteness.
- For processing-time tests, the paper uses a restricted HMDA version with application dates and action dates.
- Processing time is the number of days between the application date and the date when the loan is originated or otherwise acted on.
- The main HMDA sample runs from January 2010 through June 2016 because loans submitted later in 2016 may not have completed processing by year-end.

Interpretation: HMDA is ideal for studying processing speed and application outcomes because it records both originated and non-originated applications. It is less useful for default because HMDA does not track ex post loan performance.

2.3 Ginnie Mae FHA loan-performance data

- To study default, the paper uses Ginnie Mae mortgage-backed securities data on FHA-insured loans.
- FHA loans are important because they serve many lower-income and first-time homebuyers.
- The default outcome is whether the loan becomes 90 or more days delinquent over the observation period.
- The sample is 30-year fixed-rate FHA mortgages from September 2013 to May 2017.
- Key controls include FICO, loan-to-value ratio, debt-to-income ratio, loan purpose, and fixed effects for month and geography.

Interpretation: The default tests answer whether faster FinTech processing reflects lower screening quality. A higher default rate would support the “less careful” hypothesis; the paper finds the opposite.

2.4 CRISM data for refinancing behavior

- To study refinancing, the paper uses Equifax Credit Risk Insight Servicing McDash data.
- CRISM merges mortgage servicing records from Black Knight with borrower credit bureau information from Equifax.
- The sample tracks outstanding loans over time and identifies whether a borrower refinances.
- The analysis focuses on the 500 largest counties by loan volume from January 2010 through June 2016.
- FinTech market share is measured from HMDA, usually as a four-quarter moving average lagged one quarter.

Interpretation: CRISM allows the authors to study whether FinTech affects not just mortgage originations, but the behavior of existing mortgage borrowers who may refinance.

2.5 Refinancing incentive measure

- The paper uses the Agarwal, Driscoll, and Laibson (ADL) rule to calculate when a borrower should refinance.
- The refinancing incentive is the current mortgage rate minus the market refinance rate minus the ADL optimal refinancing differential.
- Negative incentives mean the borrower should not refinance according to ADL.
- Positive incentives mean the borrower should refinance.
- The authors bin borrowers by this incentive to distinguish efficient refinancing from excessive refinancing.

Interpretation: The refinancing analysis is not just about volume. It asks whether FinTech leads borrowers to refinance when it is privately valuable, rather than generating unnecessary refinancing churn.

2.6 Borrower access and selection variables

- The paper studies who borrows from FinTech lenders using borrower, neighborhood, and county variables.
- Access-to-finance variables include credit scores and bank branch density.
- Technology adoption variables include population density, age, education, and Internet access measures.
- Local housing-market variables include home price appreciation, local processing-time coefficients, and home prices.

Interpretation: These tests ask whether FinTech is primarily a democratizing credit-access technology or a convenience technology used by borrowers who are already financially and digitally connected.

3 Empirical specifications and interpretation

3.1 Processing-time regression

The baseline loan-level regression is

$$\text{Processing Time}_{ijct} = \delta_{ct} + \beta \text{FinTech}_j + \gamma' \text{Controls}_{ijct} + \varepsilon_{ijct}. \quad (1)$$

- i indexes loans, j lenders, c census tracts, and t application months.
- δ_{ct} are census-tract-by-month fixed effects.
- FinTech_j is one if the lender is classified as FinTech.
- Controls include borrower and loan characteristics.
- The coefficient β is the processing-time advantage of FinTech lenders, in days.

Interpretation: The fixed effects compare FinTech and non-FinTech loans within the same local market and month. A negative β means FinTech applications close faster after controlling for local demand and observable loan complexity.

3.2 Default regression

For FHA loans, the paper estimates

$$\text{Default}_{ijst} = \alpha + \beta \text{FinTech}_j + \gamma' \text{Controls}_{ijst} + \varepsilon_{ijst}. \quad (2)$$

- $\text{Default}_{ijst} = 1$ if the loan becomes 90+ days delinquent.
- j is the Ginnie Mae issuer, s is state, and t is origination month.
- Controls include origination month, state or state-by-month fixed effects, purpose fixed effects, FICO, LTV, DTI, and other loan controls.

Interpretation: If FinTech speed comes from weak screening, β should be positive. The paper finds negative coefficients, suggesting FinTech speed is not achieved by originating riskier FHA loans.

3.3 Processing-time elasticity to application volume

To test capacity constraints, the paper estimates

$$\text{Processing Time}_{ijct} = \gamma \text{Applications}_t + \beta (\text{Applications}_t \times \text{FinTech}_j) + \alpha_j + \delta_c + \theta' \text{Controls}_{it} + \varepsilon_{ijct}. \quad (3)$$

- Applications_t is log aggregate mortgage applications in month t .
- α_j are lender fixed effects.
- δ_c are census-tract fixed effects.
- β measures whether FinTech processing times are less sensitive to aggregate demand shocks.

Interpretation: Traditional lenders face processing-capacity bottlenecks when volume spikes. A negative interaction means FinTech lenders scale better under demand surges.

3.4 Denial-rate elasticity to application volume

The denial-rate regression uses the same structure as equation (3), but the dependent variable is an indicator for whether an application is denied:

$$Denied_{ijct} = \gamma Applications_t + \beta(Applications_t \times FinTech_j) + \alpha_j + \delta_c + \theta' Controls_{it} + \varepsilon_{ijct}. \quad (4)$$

Interpretation: This addresses a rationing concern. If FinTech lenders only appear scalable because they reject more borrowers when volume rises, the interaction would be positive. The paper finds it is negative.

3.5 County-level refinancing propensity

The county-level refinancing specification is

$$RefiPropensity_{ct} = \alpha_c + \lambda_t + \beta FT \text{ share}_{c,t-1} + \gamma' X_{ct} + u_{ct}. \quad (5)$$

- $RefiPropensity_{ct}$ is the share of outstanding mortgages in county c that refinance in month t .
- $FT \text{ share}_{c,t-1}$ is lagged FinTech market share, measured using HMDA.
- α_c are county fixed effects and λ_t are month fixed effects.
- Controls include average FICO, CLTV, current rate, and FHA/VA share.

Interpretation: This design compares changes in refinancing propensity within the same county over time as FinTech market share changes, controlling for aggregate refinancing cycles.

3.6 Loan-level refinancing incentive bins

The loan-level refinancing regression is estimated separately across ADL incentive bins:

$$\mathbf{1}\{Refi_{imt} = 1\} = \alpha_c + \lambda_t + \beta_b FT \text{ share}_{c,t-1} + \gamma' LoanControls_{imt} + e_{imt}, \quad (6)$$

where b indexes the refinancing-incentive bin.

Interpretation: If FinTech reduces errors of omission, the effect should be strongest when borrowers have a positive or near-positive refinancing incentive. If it causes wasteful churn, it should be strong even for negative incentives. The estimates favor the efficiency interpretation.

3.7 Borrower selection into FinTech

The borrower-selection regressions estimate a linear probability model:

$$\mathbf{1}\{Originator \text{ is } FinTech\}_{ijct} = \alpha_t + \beta' Z_{ijct} + \gamma' W_{ct} + \varepsilon_{ijct}. \quad (7)$$

- The dependent variable equals 100 if the originator is a FinTech lender and zero otherwise.
- Continuous variables are standardized to have mean zero and standard deviation one.
- Regressions are estimated separately for purchases/refinances and for all lenders/nonbanks.

Interpretation: These regressions characterize the type of borrower and local market associated with FinTech use. They do not establish causal effects, but they are informative about whether FinTech primarily serves underserved borrowers.

4 Identification and empirical design

4.1 What makes the processing-time comparison credible

- The key comparison is between FinTech and non-FinTech applications within the same census tract and month.
- Census-tract-month fixed effects absorb local housing-market conditions, local regulations, local appraisal delays, and local demand shocks.
- Loan controls absorb observable loan complexity and borrower characteristics.
- Separate estimates for purchases and refinances address the fact that refinance loans are more automatable than purchase loans.

Interpretation: The empirical design does not randomly assign borrowers to lenders. Its credibility comes from rich local-time fixed effects and the robustness of a large processing-time gap across samples and controls.

4.2 Why default tests matter

- Faster lending could reflect better technology or weaker screening.
- FHA loans provide a useful setting because loan performance is observed and the borrower segment is relatively risky.
- The paper compares FinTech and non-FinTech default rates after controlling for FICO, LTV, DTI, loan purpose, geography, and time.

Interpretation: The default evidence is a falsification-style test of the most concerning alternative mechanism. If FinTech originated carelessly screened loans, default rates should be higher.

4.3 Why aggregate application-volume shocks are useful

- Mortgage demand fluctuates sharply with refinancing incentives and interest rates.
- Aggregate application volume is plausibly not caused by any one lender's capacity choice.
- The interaction with FinTech tests whether FinTech lenders are more scalable.
- The denial-rate test distinguishes scalability from credit rationing.

Interpretation: The evidence supports the view that FinTech operations reduce capacity constraints rather than merely changing borrower composition.

4.4 Why refinancing tests are economically important

- Many borrowers refinance too slowly even when refinancing would save money.
- FinTech may lower search, paperwork, and transaction costs.
- County fixed effects absorb persistent local refinancing culture and market structure.
- Month fixed effects absorb national interest-rate/refinancing cycles.
- ADL incentive bins test whether additional refinancing is efficient.

Interpretation: The refinancing results connect FinTech lending to a real household-finance friction: failure to refinance when interest savings justify it.

4.5 Limits of the design

- FinTech adoption is not randomized across borrowers, lenders, or counties.
- Borrowers who choose online lenders may differ in unobserved ways.
- Ginnie Mae identifies issuers rather than originators, introducing measurement error in default tests.
- The paper observes outcomes through 2016/2017, before the full diffusion of later digital mortgage technologies.
- The results do not necessarily imply FinTech expands access to credit for the most underserved borrowers.

Interpretation: The paper's strongest evidence concerns speed, scalability, and refinancing frictions. The access-to-finance question is more descriptive and less causal.

5 Tables and figures

This section keeps the same *Read* plus *Economic message* structure, with adjacent original figures/tables grouped to save space. All visuals are directly cropped from the paper.

5.1 Figure 1 and Table 1: rise and identity of FinTech mortgage lenders

Read:

- Figure 1 shows that the number of nonbank FinTech mortgage lenders is small and flat through 2013, then grows rapidly after 2015.
- By 2017, the count reaches close to 20 FinTech lenders.
- Table 1 lists the largest mortgage originators in 2016 and flags FinTech originators.
- Quicken Loans is the largest FinTech lender, with about \$90.55 billion in volume and a 4.52% market share in 2016.
- LoanDepot and Guaranteed Rate are also sizable FinTech lenders; several newer FinTech lenders appear outside the top 20.

Economic message: The paper studies a real industry shift: online mortgage origination moves from niche to significant market presence during the sample.



Figure 1
Number of FinTech mortgage lenders (according to our classification) over time

(a) Figure 1: number of FinTech mortgage lenders

FinTech and top-20 mortgage originators in 2016, based on the Home Mortgage Disclosure Act (HMDA)

Rank	Type of lender	Lender name	Volume (bn)	Market share (%)	FinTech since
1	Bank	Wells Fargo	132.58	6.62	
2	FinTech	Quicken Loans	90.55	4.52	2010
3	Bank	JPMorgan Chase	75.52	3.77	
4	Bank	Bank Of America	60.24	3.01	
5	FinTech	LoanDepot.com	35.94	1.80	2016
6	Mtg	Freedom Mortgage	32.17	1.61	
7	Bank	US Bank	30.69	1.53	
8	Mtg	Caliber Home Loans	27.94	1.40	
9	Bank	Flagstar	27.31	1.36	
10	Mtg	United Shore	22.93	1.15	
11	Bank	Citibank	21.73	1.09	
12	FinTech	Guaranteed Rate	18.44	0.92	2010
13	Mtg	Finance of America	17.63	0.88	
14	Mtg	Fairway Independent	16.10	0.80	
15	Bank	USAA Federal Savings	15.52	0.78	
16	Mtg	Guild Mortgage	15.07	0.75	
17	Mtg	Stearns Lending	14.93	0.75	
18	Bank	Suntrust Mortgage	14.77	0.74	
19	Bank	Primelending	13.87	0.69	
20	Mtg	Nationstar Mortgage	13.50	0.67	
...					
23	FinTech	Movement Mortgage	11.61	0.58	2014
39	FinTech	Everett Financial (Supreme)	7.62	0.38	2016
534	FinTech	Avex Funding (Better.com)	0.49	0.02	2016

Bank = Depository Institution, Mtg = Nonbank Mortgage Lender, FinTech = FinTech Lender. Market share is based on dollar volume of originations in HMDA. This table reports all lenders classified as FinTech at the end of 2016. Our classification also identified 12 additional nonbank lenders who began offering an online application platform in the first half of 2017, after the end of our HMDA sample period: FBC Mortgage Inc.; RPM Mortgage Inc.; ARK-LA-TEX Financial Services (Benchmark Mortgage); Envoy Mortgage; Skyline Financial

(b) Table 1: top mortgage originators and FinTech lenders

5.2 Table 2: descriptive statistics by lender type

Read:

- FinTech lenders are much more refinance-focused: 78% of their originated mortgages are refinances, compared with 66% for banks and 48% for non-FinTech nonbanks.
- FinTech lenders have shorter median processing time: 37 days for originated mortgages, compared with 45 days for banks and 40 days for non-FinTech nonbanks.
- FinTech lenders have a relatively high share of FHA lending compared with banks, but lower FHA share than non-FinTech nonbanks.

- For all applications, FinTech denial rates are higher than other lender types, but the paper later shows this does not mean FinTech lenders ration more under volume shocks.

Economic message: FinTech lenders look operationally different even before regression controls: they specialize in refinances and process loans faster.

Descriptive statistics by lender type, HMDA data, 2010 - mid-2016

	Banks		Nonbank lenders				All lenders	
	Mean	p50	Non-FinTech		FinTech		Mean	p50
			Mean	p50	Mean	p50		
Originated mortgages								
Applicant income	121	86.00	102	82.00	102	84.00	115	84.00
Loan amount / income	1.96	1.80	2.46	2.40	2.34	2.19	2.13	2.00
Home purchase	0.34	0	0.52	1	0.22	0	0.38	0
Refinancing	0.66	1	0.48	0	0.78	1	0.62	1
Jumbo	0.05	0	0.02	0	0.02	0	0.04	0
Loan type:								
Conventional	0.86	1	0.61	1	0.71	1	0.78	1
FHA	0.09	0	0.28	0	0.20	0	0.15	0
VA	0.05	0	0.11	0	0.09	0	0.07	0
Owner occupied	0.88	1	0.92	1	0.92	1	0.89	1
Male	0.67	1	0.69	1	0.59	1	0.68	1
Female	0.25	0	0.27	0	0.26	0	0.26	0
No coapplicant	0.45	0	0.52	1	0.50	0	0.48	0
Race:								
White	0.79	1	0.78	1	0.68	1	0.78	1
Black/African American	0.04	0	0.06	0	0.05	0	0.05	0
Asian	0.05	0	0.07	0	0.04	0	0.06	0
Other	0.01	0	0.01	0	0.01	0	0.01	0
Unknown	0.11	0	0.09	0	0.22	0	0.11	0
Processing time	53.04	45.00	50.20	40.00	42.58	37.00	51.71	43.00
Observations	32,751,662		14,742,227		2,306,237		49,800,126	
All applications								
Loan outcome								
Originated	0.64	1	0.58	1	0.66	1	0.62	1
Approved, not accepted	0.04	0	0.05	0	0.03	0	0.04	0
Denied	0.20	0	0.16	0	0.27	0	0.19	0
Withdrawn	0.09	0	0.15	0	0.03	0	0.11	0
Closed for incompleteness	0.03	0	0.06	0	0.01	0	0.04	0
Processing time	47.16	40.00	46.98	35.00	40.11	35.00	46.80	38.00
Observations	51,448,444		25,604,501		3,473,506		80,526,451	

This table contains summary statistics of HMDA data by lender type, for loan applications from January 2010 through June 2016. "Banks" are depository institutions, "Nonbank lenders" are nonbank mortgage lenders, and "FinTech" lenders are classified according to Section 2. In the first part of the table, summary statistics are calculated for originated mortgages only. In the second part of the table, statistics are calculated for all applications, which include applications that ended up being originated, approved by the lender but not accepted by the borrower, denied, withdrawn by the applicant before a decision was made, or closed for incompleteness.

Table 2: descriptive statistics by lender type

5.3 Table 3: loan-level processing-time results

Read:

- For purchase loans, FinTech lenders process applications about 7.45 to 9.43 days faster across specifications.
- For refinance loans, FinTech lenders process applications about 9.20 to 14.75 days faster.
- The largest refinance estimate in Column 4 is -14.75 days, relative to a mean processing time of about 51 days.
- The effect persists with census-tract-month fixed effects, loan controls, and a nonbank-only comparison sample.

Economic message: FinTech speed is not explained by local geography, timing, or basic borrower/loan characteristics. The advantage is especially large for refinancing, where online automation is more useful.

FinTech lenders and processing times: Loan-level results					
<i>A. Purchase loans</i>	(1)	(2)	(3)	(4)	(5)
FinTech lender	-7.93*** (2.22)	-9.43*** (2.84)	-8.31*** (1.79)	-9.20*** (2.30)	-7.45*** (1.65)
log(applicant income)		-0.55*** (0.20)		-0.81*** (0.11)	-0.18 (0.24)
log(loan amount)		4.47*** (0.52)		5.22*** (0.35)	7.07*** (0.96)
FHA		0.65 (0.44)		-0.11 (0.33)	-0.96* (0.53)
VA		1.68 (1.26)		1.25 (1.05)	1.26 (1.50)
Jumbo		3.17*** (1.23)		5.32*** (0.76)	6.18*** (1.49)
Observations	19,159,345	19,159,345	18,146,785	18,146,785	6,841,626
R ²	.00	.02	.26	.28	.36
Census tract-month	No	No	Yes	Yes	Yes
Loan controls	No	Yes	No	Yes	Yes
Sample	All lenders	All lenders	All lenders	All lenders	Nonbanks
<i>B. Refinance loans</i>	(1)	(2)	(3)	(4)	(5)
FinTech lender	-9.99*** (2.46)	-13.64*** (2.35)	-11.17*** (2.08)	-14.75*** (2.20)	-9.20*** (1.93)
log(applicant income)		0.04 (0.34)		-0.16 (0.26)	-0.23 (0.36)
log(loan amount)		4.74*** (0.50)		4.32*** (0.47)	1.62** (0.67)
FHA		5.83*** (1.98)		5.40*** (1.78)	5.20*** (1.59)
VA		1.70 (2.60)		2.03 (2.16)	1.53 (2.28)
Jumbo		7.06*** (1.01)		7.32*** (0.94)	9.40*** (0.52)
Observations	30,616,247	30,616,247	29,843,262	29,843,262	7,674,028
R ²	.01	.10	.21	.27	.32
Census tract-month	No	No	Yes	Yes	Yes
Loan controls	No	Yes	No	Yes	Yes
Sample	All lenders	All lenders	All lenders	All lenders	Nonbanks

This table reports regressions of loan processing time (in days) on an indicator variable identifying FinTech lenders, census tract-month fixed effects, loan controls and borrower controls. Data source: HMDA. The sample consists of originated purchase loans in panel A and refinancing loans in panel B with application dates from 2010 to 2016Q2. Displayed loan controls include the log of applicant income, the log of the loan amount, indicators for FHA loans, VA loans and jumbo loans. Suppressed loan controls include applicant race, gender, whether the loan has a coapplicant, whether a preapproval was obtained, the occupancy and lien status of the loan, the property type, and a dummy indicating whether income is missing. Column 5 excludes bank lenders. Standard errors reported in parentheses are clustered by lender. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 3: FinTech lenders and processing times

5.4 Tables 4 and 5: default risk and screening quality

Read:

- Table 4 reports percent differences in FHA default rates between FinTech and non-FinTech loans. The estimates are generally negative.
- In the all-FHA state-level comparison, FinTech default rates are roughly 35% lower for one-year and two-year default definitions.
- Table 5 uses loan-level Ginnie Mae data with controls. FinTech loans are 0.79 to 1.51 percentage points less likely to default in the main specifications.
- Column 3 of Table 5 controls for FICO, LTV, and DTI and still finds a -0.93 percentage point coefficient.

Economic message: Faster processing does not appear to come from careless underwriting. If anything, observed FinTech FHA loans perform better than non-FinTech FHA loans.

Default rate for FinTech lenders, FHA loans (% deviation from region)

	Default definition		
	1-year default	2-year default	Mix-adjusted 2-year default
<i>A. All FHA loans</i>			
State level	-35.0*** (3.8)	-35.8*** (3.9)	-25.5*** (4.5)
MSA level	-35.3*** (2.4)	-36.2*** (2.6)	
<i>B. High market share regions only (loans in top quartile markets by lender):</i>			
State level	-24.7*** (8.4)	-23.4*** (8.6)	-11.7 (9.9)
MSA level	-19.8*** (6.0)	-20.5*** (6.1)	
<i>C. Disaggregated by purchase versus refinancing (all loans, state level)</i>			
Purchase	-14.8*** (3.6)	-17.1*** (4.3)	-10.1* (5.9)
Refinancing	-30.6*** (2.6)	-27.5*** (3.5)	-40.3*** (2.4)
<i>D. Disaggregated by neighborhood socioeconomic status (all loans, state level)</i>			
Underserved (low income/minority)	-33.5*** (4.5)	-32.4*** (4.4)	-25.3*** (4.4)
Not underserved	-36.8*** (3.6)	-36.5*** (3.9)	-25.4*** (4.3)
<i>E. All FHA loans: longer time series</i>			
National level	-44.7*** (4.8)	-45.6*** (4.5)	-32.7*** (5.7)
State level	-45.4*** (3.2)	-46.1*** (3.1)	-33.4*** (3.9)

This table reports weighted average percent difference in default rate between mortgages from FinTech lenders and all FHA mortgages originated in same time period and market (MSA, state, or national market). Values less than zero indicate lower default rates for FinTech lenders. These statistics are calculated as the weighted average of $compare_{it} = (default_{it} / default_{it}) - 1$ across FinTech lenders i and regions r , weighting by lender origination volume. In practice we calculate this weighted average by regressing $compare_{it}$ on a constant term using weighted least squares. Default definition is default within first year, default within first 2 years, or the "mix-adjusted" default rate, which is based on the supplementary performance metric, an adjusted default rate, which takes into account the credit score distribution of originations (see text for details). Sample period 2015:Q3 to 2017:Q3, except for panel E, where sample period is 2012:Q3 to 2017:Q3. Data are extracted from the FHA NW Early Warning System portal in December 2017. Robust standard errors in parentheses; standard errors are clustered by state in a state-level regression using the longer time series from E. * $p < .1$; ** $p < .05$; *** $p < .01$.

(a) Table 4: FHA default rate comparisons

FHA mortgage default regressions based on Ginnie Mae data

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
FinTech	-1.29*** (0.33)	-0.97*** (0.30)	-0.93*** (0.27)	-1.51*** (0.46)	-0.79*** (0.16)		-0.91*** (0.29)
FinTech share						-0.10 (1.43)	
FT $\times$ FT share							-1.70 (2.05)
FICO			-0.04*** (0.00)	-0.05*** (0.00)	-0.03*** (0.00)	-0.04*** (0.00)	-0.04*** (0.00)
LTV			0.04*** (0.01)	0.07*** (0.01)	0.03*** (0.01)	0.04*** (0.01)	0.04*** (0.01)
DTI			0.06*** (0.01)	0.07*** (0.01)	0.04*** (0.01)	0.06*** (0.01)	0.06*** (0.01)
Purpose FEs	No	Yes	Yes	Yes	Yes	Yes	Yes
Month FEs	Yes	Yes	No	No	No	Yes	No
State FEs	No	No	No	No	No	Yes	No
Month $\times$ state FEs	No	No	Yes	Yes	Yes	No	Yes
Loan controls	No	No	Yes	Yes	Yes	Yes	Yes
Mean dep. var.	3.65	3.65	3.65	4.00	2.73	3.65	3.65
R^2	.02	.02	.04	.05	.03	.04	.04
Observations	4,097,569	4,097,568	4,097,544	2,966,644	1,130,881	4,097,548	4,097,544
Loan sample	All	All	All	Purch.	Refi	All	All

Table reports regressions of indicator for a loan ever entering 90+ day delinquency on an indicator variable identifying FinTech issuers (or the state-level FinTech market share, demeaned by month; or the interaction of the FinTech indicator with FinTech market share), state-by-origination month fixed effects, loan controls and borrower controls. Coefficients are expressed in percentage points. The sample consists of FHA-insured 30-year fixed-rate mortgages originated over June 2013 to May 2017, obtained from Ginnie Mae MBS monthly loan-level disclosures. Displayed loan controls include the borrower FICO score, the loan-to-value (LTV) ratio and the debt-to-income (DTI) ratio. Suppressed loan controls include loan purpose type, the log of the loan amount, and indicators for the number of borrowers, the property type, whether the borrower received down payment assistance, and for whether FICO, LTV, or DTI are missing. Standard errors reported in parentheses are clustered by issuer. * $p < .1$; ** $p < .05$; *** $p < .01$.

(b) Table 5: Ginnie Mae default regressions

5.5 Figure 2 and Table 6: demand shocks and processing capacity

Read:

- Figure 2 shows that application volume rises and falls with refinancing incentives, and that processing times move with application volume.
- Table 6 estimates processing-time sensitivity to aggregate application volume.
- The interaction $\ln(app. vol.) \times FinTech$ is negative in all columns.
- In Column 2, a 10% increase in application volume raises processing time for non-FinTech lenders by about 1.34 days, but the FinTech interaction reduces this by about 0.61 days.
- The interaction is especially large for refinances, consistent with FinTech specialization.

Economic message: FinTech lending scales better when mortgage demand surges. Technology reduces capacity constraints, rather than merely lowering average processing time.

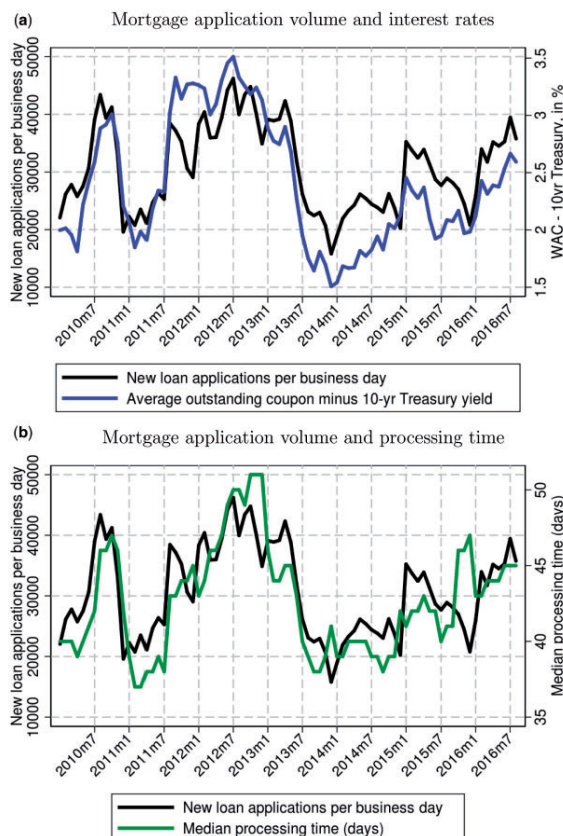


Figure 2
Application volume, interest rate, and processing time

(a) Figure 2: application volume, rates, and processing time

Elasticity of processing time with respect to aggregate application volume: FinTech versus other lenders

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
ln(app. vol.)	11.76*** (1.37)	13.42*** (1.52)	18.79*** (2.30)	13.38*** (1.34)	8.88*** (1.46)	13.60*** (2.80)	10.55*** (2.14)
ln(app. vol.) × FinTech	-7.55*** (2.17)	-6.12*** (1.81)	-9.51*** (2.19)	-7.45*** (2.07)	-2.08 (1.60)	-4.45** (2.25)	-4.48* (2.35)
Observations	49,775,550	49,774,855	30,615,102	80,495,607	17,023,266	8,925,769	29,047,814
R ²	.14	.20	.25	.17	.21	.17	.17
Loan controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Lender FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Census tract FEs	No	Yes	Yes	Yes	Yes	Yes	Yes
Month FEs	No	Yes	Yes	Yes	Yes	Yes	Yes
Application sample	Originated	Originated	Refi	All	Originated	Refi	All
Lender sample	All	All	All	All	Nonbanks	Nonbanks	Nonbanks

Table reports regressions of loan processing time (in days) on the log of aggregate monthly application volume, an interaction with the FinTech indicator, loan controls, lender fixed effects, census-tract fixed effects, and calendar month fixed effects. Data source: HMDA. The sample is restricted to application dates from 2010 to 2016:Q2. Columns 1, 2, and 5 include all originated loans; Columns 3 and 6 include originated refinancing loans; and Columns 4 and 7 include all applications (including denied and withdrawn applications). The sample of lenders includes all lender types in Columns 1–4 and nonbanks only in Columns 5–7. Loan controls include the log of applicant income, the log of the loan amount, indicators for FHA loans, VA loans, and jumbo loans, applicant race, gender, whether the loan has a coapplicant, whether a preapproval was obtained, the occupancy and lien status of the loan, the property type, and a dummy indicating whether income is missing. Columns 4 and 7 also include indicators for whether a loan was denied or withdrawn. Standard errors reported in parentheses are clustered by lender. * $p < .1$; ** $p < .05$; *** $p < .01$.

(b) Table 6: processing-time elasticity

5.6 Figure 3, Table 7, and Table 8: elastic supply versus rationing

Read:

- Figure 3 visually shows that FinTech processing times are lower and have a flatter slope with respect to aggregate application volume.
- Table 7 shows that the interaction of application volume with FinTech is negative for denial probabilities.
- Thus, FinTech lenders do not maintain speed by denying more applications when volume is high.
- Table 8 shows no strong evidence that FinTech origination volume responds less to application-volume changes than other lenders.

Economic message: FinTech lenders appear to supply credit more elastically. They process faster under load and do not simply ration borrowers away.

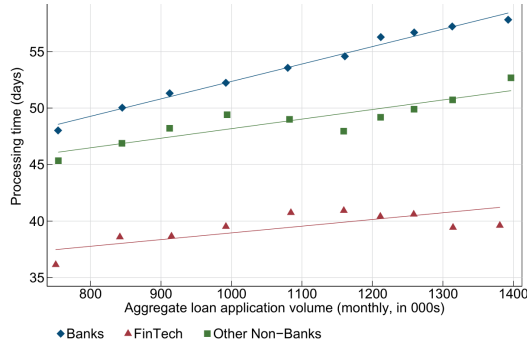


Figure 3
Illustrating differences in processing times and elasticity to demand across lender types
 This figure shows binned scatter plot (with linear fit) of processing times of originated loans against the volume of aggregate loan applications by lender type, all measured in HMDA 2010-2016:Q2. Processing times are

Elasticity of mortgage application denial probabilities with respect to variation in aggregate demand for loans: FinTech versus other lenders

	(1)	(2)	(3)	(4)
ln(app. vol.)	-0.060*** (0.009)	-0.108*** (0.011)	-0.066*** (0.017)	-0.123*** (0.021)
ln(app. vol.) × FinTech	-0.108*** (0.018)	-0.087*** (0.011)	-0.109*** (0.021)	-0.072*** (0.021)
Loan controls	Yes	Yes	Yes	Yes
Lender FEs	Yes	Yes	Yes	Yes
County tract FEs	Yes	Yes	Yes	Yes
Observations	68,793,029	44,727,774	23,537,046	13,327,482
Application sample	All	Ref	All	Ref
Lender sample	All	All	Nonbanks	Nonbanks

Table reports regressions of indicator for loan application denial on the log of aggregate application volume, an interaction with the FinTech indicator, loan controls, lender fixed effects, county tract fixed effects, and calendar month fixed effects. Data source: HMDA. The sample is restricted to application dates from 2010 to 2016:Q2. Applications are included if they result in a loan origination, in the application being approved by the lender, but not accepted by the borrower, or an application denial. The sample of lenders includes all lender types in Columns 1 and 2 and nonbanks only in Columns 3 and 4. Loan controls include the log of applicant income, the log of the loan amount, indicators for FHA loan, VA loan, and jumbo loans, applicant race, gender, whether the loan has a co-borrower, whether a prepayment was obtained, the occupancy and loan status of the loan, the property type, and a dummy indicating whether income is missing. Standard errors reported in parentheses are clustered by lender. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Elasticity of originations with respect to changes in aggregate volume: FinTech versus other lenders

	(1)	(2)	(3)	(4)
Δ ln(app. vol.)	1.147*** (0.01)	1.367*** (0.01)	1.127*** (0.02)	2.077*** (0.02)
Δ ln(app. vol.) × FinTech	0.02 (0.08)	0.06 (0.13)	0.02 (0.09)	-0.08 (0.13)
Observations	52,430	51,311	24,489	23,831
R ²	.36	.39	.33	.39
Month FEs	Yes	Yes	Yes	Yes
Application sample	Originated	Ref	Originated	Ref
Lender sample	All	All	Nonbanks	Nonbanks

Table reports regressions of the log change in lender-level originated loans on the log change in aggregate application volumes, an interaction with the FinTech indicator, lender fixed effects, and calendar month fixed effects. The unit of observation is lender-month. Data source: HMDA. The sample is restricted to 2010 through 2016:Q2. Columns 1 and 2 include all originated loans. Columns 3 and 4 include originated refinancing loans. The sample of lenders includes all lender types in Columns 1 and 2 and nonbanks only in Columns 3 and 4. Standard errors reported in parentheses are White-Huber standard errors. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

(a) Figure 3: processing-time slopes by lender type

(b) Table 7: denial elasticity

(c) Table 8: origination elasticity

5.7 Table 9 and Figure 4: FinTech market share and refinancing propensity

Read:

- Table 9 shows a positive relationship between county FinTech market share and monthly refinancing propensity.
- With controls, the coefficient is 0.689 for all mortgages and 0.706 for 30-year fixed-rate mortgages.
- The paper interprets an 8-percentage-point increase in FinTech share as raising refinancing propensity by about 0.055 percentage points, roughly 10% relative to the mean.
- Figure 4 shows that counties in the highest FinTech tercile initially refinance slowly but catch up over time.

Economic message: FinTech seems to reduce refinancing inertia. Its growth is associated with more refinancing in places that previously lagged.

FinTech market share and refinancing propensities: County-level results

	(1) All	(2) All	(3) 30yr FRM	(4) 30yr FRM
FinTech share Q_{-1} (MA)	1.121*** (0.204)	0.689*** (0.142)	1.195*** (0.223)	0.706*** (0.157)
Average FICO/10		0.067*** (0.012)		0.071*** (0.013)
Average CLTV/10		-0.094*** (0.007)		-0.104*** (0.008)
Average current rate		1.135*** (0.059)		1.202*** (0.062)
FHA/VA share		0.190 (0.315)		0.185 (0.332)
County fixed effects	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes
Mean of dep. var.	0.54	0.54	0.59	0.59
R ²	.78	.81	.76	.79
Obs.	39,000	39,000	39,000	39,000

This table reports regressions of county-level monthly refinancing propensities (defined as the share of outstanding mortgages in month $t-1$ that are refinanced in month t , in percentage points) on the FinTech share in a county (4-quarter rolling average, lagged one quarter; range [0,1]), county fixed effects, month fixed effects, and average characteristics of outstanding loans in the county: FICO, updated combined loan-to-value ratios (CLTV), average coupon rate, and the share of FHA/VA mortgages. Data sources: CRISM for refinancing propensities and loan characteristics, HMDA for FinTech market shares. In Columns 1 and 2, refinancing propensities are calculated based on all loans; in Columns 3 and 4, based on 30-year fixed-rate mortgages only. Sample covers January 2010 through June 2016 for the largest 500 counties by count of outstanding mortgages in December 2013 (see Internet Appendix E for details). Standard errors reported in parentheses are clustered by county. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

(a) Table 9: county-level refinancing propensity

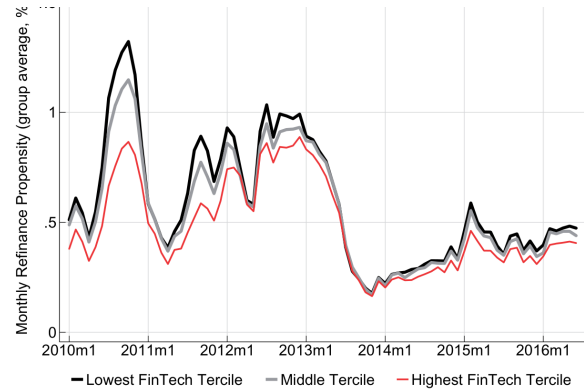


Figure 4
Refinancing propensities across counties with different levels of FinTech market share
 This figure shows average monthly refinance propensities across three groups of counties, sorted based on county-level FinTech market shares (among refinance loans) over mid-2012 to mid-2013. Data source: CRISM

(b) Figure 4: refinancing propensities by FinTech share

5.8 Table 10: refinancing optimality by incentive bins

Read:

- Table 10 separates borrowers into refinancing-incentive bins using the ADL rule.
- For borrowers more than 1 percentage point out of the money, the FinTech-share effect is slightly negative.
- The effect becomes positive and strongest when borrowers are near the optimal refinancing threshold: coefficients are about 2.01 and 1.99 in the $[-0.5, 0)$ and $[0, 0.5)$ bins.
- The pooled coefficient is 1.436.

Economic message: FinTech does not simply cause unnecessary refinancing. The additional refinancing is concentrated where borrowers are close to or above the threshold at which refinancing is economically reasonable.

FinTech market share and refinancing propensities, by refinancing incentive bins							
Refi incentive (ADL)	(1) < -1	(2) [-1, -0.5)	(3) [-0.5, 0)	(4) [0, 0.5)	(5) [0.5, 1)	(6) ≥ 1	(7) All
FT share Q_{-1} (MA)	-0.140* (0.073)	1.028*** (0.200)	2.008*** (0.304)	1.985*** (0.353)	1.444*** (0.347)	0.507* (0.267)	1.436*** (0.229)
County FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Loan controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean of dep. var.	0.12	0.46	0.85	1.04	1.05	0.78	0.59
R^2	.00	.00	.01	.01	.01	.01	.00
Obs.	64,866,392	42,085,823	38,988,748	29,249,088	19,039,098	20,745,039	214,996,787

Table reports regressions of indicator for whether a borrower refinanced their loan in a given month on the FinTech share in a county (4-quarter rolling average, lagged one quarter), county fixed effects, month fixed effects, and the following loan controls: 5-point bins of CLTV, 20-point bins of FICO, a cubic function in the age of the refinanced loan, the log of the balance of the refinanced loan, and an indicator for whether the refinanced loan was an FHA/VA loan. Data sources: CRISM for refinancing propensities and loan characteristics, HMDA for FinTech market shares. For Columns 1–6, borrowers are separated into six bins depending on their refinancing incentive based on the Agarwal, Driscoll, and Laibson (2013) (ADL) calculation. Negative incentives (expressed in percentage points of interest rates) mean that according to ADL a borrower should not refinance; positive incentives mean they should refinance. Column 7 pools all bins. Sample includes 30-year fixed-rate mortgages only. Standard errors reported in parentheses are clustered by county. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 10: refinancing propensity by ADL incentive bins

5.9 Table 11: who borrows from FinTech mortgage lenders?

Read:

- The dependent variable equals 100 if the originator is a FinTech lender.
- FinTech use is higher among some groups consistent with online adoption: more educated areas, higher population density, and younger borrowers in several specifications.
- The evidence does not strongly support an access-to-finance story. For example, borrowers with lower credit scores are not consistently more likely to use FinTech once comparisons are restricted to nonbanks.
- Bank branch density is positively associated with FinTech use in several columns, which is inconsistent with the view that FinTech mainly serves bank-branch deserts.
- Local processing-time coefficients are positively related to FinTech use, suggesting FinTech grows where traditional lending is slower.

Economic message: FinTech adoption looks more like a convenience and technology-adoption channel than a broad expansion of credit access to underserved borrowers.

Who borrows from FinTech mortgage lenders?

Dependent variable: = 100 if originator is FinTech lender, = 0 otherwise

	Purchases		Refinances	
	All	Nonbanks	All	Nonbanks
<i>Borrower income and demography</i>				
log(income)	0.104*** (0.00650)	0.701*** (0.0173)	-0.833*** (0.00725)	-0.159*** (0.0275)
Gender:				
Female	0.0592*** (0.00947)	0.184*** (0.0208)	0.756*** (0.0126)	3.056*** (0.0379)
Unknown	2.887*** (0.0421)	10.13*** (0.117)	6.728*** (0.0437)	24.99*** (0.100)
Race and ethnicity:				
Black	-0.306*** (0.0254)	-0.387*** (0.0495)	-0.415*** (0.0291)	1.166*** (0.0814)
Hispanic	-0.880*** (0.0200)	-1.577*** (0.0391)	-1.432*** (0.0250)	-1.982*** (0.0629)
Unknown	1.551*** (0.0294)	3.220*** (0.0658)	3.632*** (0.0310)	6.540*** (0.0710)
% black or hispanic ^{TRACT}	-0.228*** (0.0166)	-1.064*** (0.0394)	-0.256*** (0.0165)	-2.273*** (0.0501)
<i>Access to finance</i>				
Credit score ^{TRACT}	-0.279*** (0.0192)	-0.731*** (0.0468)	-1.068*** (0.0193)	-3.002*** (0.0618)
Bank branch density ^{TRACT}	0.467*** (0.0262)	0.954*** (0.0574)	0.275*** (0.0201)	0.479*** (0.0530)
<i>Technology diffusion and adoption</i>				
Population density ^{TRACT}	0.141*** (0.0275)	0.920*** (0.0697)	-0.0691*** (0.0236)	0.421*** (0.0607)
Borrower age ^{TRACT}	0.119*** (0.0168)	0.340*** (0.0400)	0.263*** (0.0169)	0.869*** (0.0502)
% bachelor degree ^{TRACT}	0.307*** (0.0213)	0.920*** (0.0529)	0.262*** (0.0180)	0.690*** (0.0553)
<i>Internet access</i>				
% high speed coverage ^{TRACT}	0.101*** (0.0127)	0.255*** (0.0316)	0.0689*** (0.0127)	0.371*** (0.0461)
% with broadband subscription ^{CTY}	-0.132*** (0.0179)	-0.487*** (0.0460)	-0.0344** (0.0167)	-0.0551 (0.0555)
<i>Local housing market conditions</i>				
% home price appreciation ^{CTY}	-0.0362*** (0.0114)	-0.836*** (0.0258)	0.277*** (0.0132)	-1.258*** (0.0382)
Processing time coefficients ^{TRACT}	0.0182 (0.0111)	0.205*** (0.0269)	0.588*** (0.0119)	1.599*** (0.0397)
log(2010 home price) ^{CTY}	-0.127*** (0.0188)	-0.688*** (0.0471)	-0.812*** (0.0213)	-2.993*** (0.0675)
Additional controls	Yes	Yes	Yes	Yes
Observations	20,790,255	8,901,875	32,936,746	9,888,845
Mean of dependent variable	2.888	6.745	6.129	20.41

Linear probability model based on HMDA data from 2010-16. All continuous right-hand size variables normalized to have mean of zero and standard deviation of one. Superscripts ^{TRACT} and ^{CTY} indicate variable is measured at the census tract or county level of aggregation, respectively, rather than at the loan level. Robust standard errors in parentheses, clustered by census tract. Regressions include year-month dummies as well as controls for loan size, loan type, dummies for jumbo loan, coapplicant, owner occupied, other race categories, and missing values

Table 11: borrower selection into FinTech lenders

6 Short summary

- The paper studies whether technology-based mortgage lenders improve the U.S. mortgage market.
- FinTech lenders are defined by the ability to complete the mortgage application fully online.
- FinTech market share rises from about 2% to 8% from 2010 to 2016.
- FinTech lenders process applications about 20% faster than other lenders.
- The speed advantage is larger for refinances than purchases.
- Faster processing is not associated with higher default rates.
- In FHA data, FinTech loans default less often in the paper’s main specifications.
- FinTech processing times are less sensitive to aggregate application-volume shocks.
- FinTech lenders do not appear to maintain speed by rejecting more applicants.
- Counties with greater FinTech market share refinance more.
- Additional refinancing is concentrated among borrowers for whom refinancing is financially sensible.
- The paper finds little evidence that FinTech lenders primarily target borrowers with low access to finance.
- The main economic channel is reduced processing and transaction frictions, not necessarily broader credit democratization.

7 Conclusion

7.1 Core conclusion

Fuster, Plosser, Schnabl, and Vickery show that mortgage FinTech changes the production technology of lending. The strongest evidence is that FinTech lenders process loans faster, scale better when demand surges, and help borrowers refinance more efficiently. The paper does not find evidence that this speed comes at the expense of higher default risk.

7.2 Economic intuition

Mortgage origination involves documentation, verification, underwriting, communication, and closing. Online applications and centralized processing reduce frictions in these steps. Lower frictions shorten processing times and make lender capacity more elastic. For refinance borrowers, lower transaction costs also reduce inertia and make it more likely that borrowers refinance when doing so saves money.

7.3 Takeaway

The paper’s main message is that financial technology can improve the efficiency of lending markets without necessarily expanding credit to underserved borrowers. FinTech mortgage lending is valuable because it lowers operational frictions, not because it fully solves access-to-credit problems.